ChatGPT

Insurance Fraud Factors and Data Model

Fraud Factors and Red Flags by Insurance Category

Insurance fraud often involves exaggerating or falsifying claim details. Below we outline common fraud subcategories in health, auto, and property insurance, along with key **fraud indicators** (red flags) for each and the rationale behind them.

Health Insurance Fraud

- **Upcoding:** Billing for a more expensive or higher-level service than was actually provided [1]. This inflates claim payouts fraudulently.
- **Unusually High Procedure Codes:** The provider frequently uses the highest-severity or most expensive billing codes for routine treatments. *Rationale:* Upcoding is indicated when a provider's coding pattern skews toward costlier procedures not justified by the diagnoses [1]. An atypically high rate of expensive codes compared to peers suggests the provider may be misrepresenting service complexity for profit.
- **Diagnosis–Treatment Mismatch:** The treatment intensity or duration billed is inconsistent with the patient's documented condition. *Rationale:* If minor diagnoses (e.g. common cold) routinely correspond to high-cost procedures, it flags possible upcoding. Legitimate care should align with medically necessary services; discrepancies may mean the provider billed for a more complex service than needed.
- **Lack of Supporting Documentation:** Medical records or visit notes do not substantiate the level of service billed. *Rationale:* High-level codes (e.g. extended consultations, complex surgeries) require extensive documentation. Inadequate evidence for a pricey procedure is a red flag that the billing code was inflated.
- **Phantom Billing:** Charging for medical services, tests, or supplies that were never provided to the patient [1].
- **No Corresponding Patient Visit:** A claim for a service date where the patient has no appointment or medical record entry. *Rationale:* If a bill is submitted for treatment on a day when the patient wasn't seen or treated, it suggests the provider invented a claim (phantom service) [2]. Legitimate services always leave some record (appointment logs, clinical notes).
- **Duplicate Claims for One Service:** Submitting the same procedure claim multiple times (so-called **double billing** [1]). *Rationale:* If two or more claims have identical patient info, dates, and procedures, it indicates the provider is trying to be paid twice for one service. Robust systems check for duplicate claim IDs or service dates to catch this.
- **Patient Complaints of Unknown Charges:** Patients report insurance statements listing treatments they never received. *Rationale:* Phantom billing is often exposed when patients review their Explanation of Benefits and see charges for phantom services. A pattern of such complaints against one provider is a serious red flag.

- **Unbundling:** Splitting a procedure into multiple smaller billing codes that should be billed as one package [1]. This exaggerates the claim by inflating the total cost.

- **Multiple Related Codes Instead of One:** The claim contains several separate procedure codes for what is clinically a single combined procedure. *Rationale:* For example, billing an evaluation, procedure, and follow-up care separately when a single bundled code exists for the treatment. Unbundling maximizes reimbursement by exploiting coding rules [3]. An audit flag is raised if a provider persistently avoids bundled codes in favor of itemized billing.
- **High Frequency of Component Billing:** The provider often bills component services individually across many patients. *Rationale:* Occasional separate billing might be innocent error, but a consistent pattern suggests intentional unbundling. Rule-based systems can flag providers with statistically higher unbundled billing than the norm for similar cases.

- **Unusual Code Combinations:** Certain procedure code combinations that are medically unlikely to occur together except as part of one overall treatment. *Rationale:* Coding guidelines (like the NCCI in Medicare) define which services are typically bundled [4]. Bills breaking these norms (e.g. an incision and the closure billed separately) signal potential fraud.

- **Performing Unnecessary Services:** Providing and billing for tests or treatments that the patient doesn't need, purely to increase billing.

- **Excessive or Frequent Testing:** Repeated diagnostics or lab tests for patients without clear clinical need. *Rationale:* If a provider orders an unusually high number of expensive tests or procedures (MRI, labs, etc.) for mild conditions, it may indicate they are padding the claim. Performing unnecessary services "solely for financial gain" is a known fraud tactic [5].
- **Falsified Diagnoses to Justify Treatment:** Diagnosing a patient with a condition they do not have (or exaggerating its severity) to bill insurance for more services. *Rationale:* If claims data shows certain providers consistently coding serious diagnoses that differ from prior patient history or second opinions, this suggests the diagnosis was inflated to authorize costly procedures [6]. Cross-checking patient records or sudden spikes in rare conditions per provider can reveal this pattern.

- **High Cost per Patient:** The average claim amount per patient for a provider is abnormally high. *Rationale:* Unnecessary procedures will inflate the total cost of claims. If one clinic's patients always incur far higher charges than peer clinics for similar demographics, it flags potential overtreatment or billing of phantom add-on services.

- **Identity/Insurance Misuse:** (Patient-side fraud) Using someone else's insurance or giving out your insurance info for fraudulent billing [7]. While not "claim exaggeration" by providers, it affects claims.

- **Multiple Patients Using Same ID:** Different claims (different names or locations) all using the same insurance policy number. *Rationale:* This suggests identity swapping or misuse of insurance credentials [7]. An insured individual might be "lending" their card or an identity thief is using stolen information to file claims. Systems can flag one policy number generating claims from distant clinics or simultaneous treatments.
- **Impersonated Provider or Facility:** Bills coming from unlicensed individuals posing as doctors or fake clinics. *Rationale:* If credential checks show the billing provider ID does not correspond to a

legitimate licensed practitioner (or the facility address is bogus), the entire claim may be fabricated [8]. Verification of provider licensure and existence is a key control to catch this form of fraud.

Auto Insurance Fraud

- **Inflated Repair Costs:** Exaggerating or fabricating the cost of vehicle repairs after an accident. This often involves collusion with a repair shop to overstate damages.
- **Single Shop, No Second Estimate:** The claimant insists on using a particular body shop and provides only one repair estimate, which is unusually high. *Rationale:* Fraud rings may work with specific garages that inflate quotes. Legitimate claims usually welcome multiple repair estimates or inspections. An **inflated repair bill** well above industry standard for the damage type is a red flag [9] [10].
- **Repairs Unrelated to Accident:** The itemized estimate includes parts or damage on the vehicle that are not consistent with the accident's cause or location. *Rationale:* Claimants may add pre-existing dents or unrelated fixes (e.g. old bumper damage) to the claim. If the **damage descriptions are inconsistent** with the accident (e.g., "hail damage" on a collision claim), it indicates padding the claim.

- **Maxing Out Policy Limits:** The repair quote comes in just under the policy's coverage limit or exactly matches a coverage cap. *Rationale:* While it could be coincidence, fraudsters often ensure the claim value is high but still collectible. A perfectly maxed claim suggests the cost was tailored for payout, not reflecting actual damage. Insurers often flag cases where **claimed losses equal the insured limit** as potentially inflated claims.

- **Staged Accidents:** Deliberately causing an accident or making it appear an accident occurred, in order to file claims. Organized groups may orchestrate collisions and injuries (e.g. the "swoop-and-squat" maneuver) [11].

- **Inconsistent Accident Details:** The physical evidence and witness statements don't match the claimant's story. *Rationale:* In staged crashes, damage on vehicles may show strange patterns (e.g. prior damage or mismatched impact angles) and participants' accounts often have discrepancies. Investigators look for **inconsistencies in the facts of loss** – if the damage mechanism described is not supported by how the vehicles are actually damaged, the accident may have been intentional [12].
- **Group of Claimants with Linked Histories:** The drivers and passengers in the accident often have a history of claiming together or are connected (friends, relatives, or even previous claims in common). *Rationale:* Fraud rings reuse the same people. If multiple claimants use the same address, attorney, or medical provider, or have filed claims together before, it suggests a staged scheme. Pattern analysis can reveal clusters of claims involving the same associates.
- **Accident Soon After Policy Inception:** The collision occurs shortly after the policy was purchased or coverage was increased. *Rationale:* Stagers often secure a new policy knowing they plan to make a claim [13]. A **recent policy purchase followed quickly by a large claim** is a red flag across insurance types [14]. While not proof by itself, it warrants closer investigation of the loss's legitimacy.

- **Minor Impact, Major Injury Claims:** The damage to vehicles is slight (e.g. low-speed bump) but the occupants report serious injuries. *Rationale:* In staged accidents like a "panic stop," fraudsters might inflict injuries on themselves beforehand or simply fake soft-tissue injuries and whiplash [15]. **Low vehicle damage coupled with high medical claims** is a classic red flag – genuine accidents

generally show a correlation between impact severity and injuries. (This ties into exaggerated injury fraud below.)

- **Exaggerated Injury Claims:** Claiming injuries (or exaggerating their severity/duration) from an auto accident to obtain a larger settlement (medical bills, pain and suffering, lost wages).

- **Subjective Soft-Tissue Injuries:** The claimant reports injuries like neck/back pain that are hard to medically verify, or symptoms that seem overly severe for a mild accident. *Rationale:* Sprains and whiplash are common in fraudulent injury claims because they don't show up on scans. A pattern of claims with primarily soft-tissue injury diagnoses, especially when vehicle damage was minimal, signals possible exaggeration. Insurers look for medical consistency – if a claimant's injuries far exceed what even catastrophic crashes produce, something is off [16].
- **Extended Treatment Duration:** The medical treatment (chiropractic visits, physical therapy, etc.) continues much longer than typical for the injury type. *Rationale:* Over-treating or staying off work excessively suggests the person is prolonging recovery to increase claim value. For example, months of therapy for a minor strain is a red flag. **Sudden increase in medical services** or excessive utilization for a given injury can indicate the claimant (and possibly a medical provider in collusion) is milking the claim.
- **Same Clinic/Lawyer for Many Claimants:** The injury claims are handled by a few specific medical providers or attorneys known for aggressive claims. *Rationale:* Fraudulent injury rings often involve certain doctors or lawyers who specialize in generating big claims (through inflated medical reports or legal pressure). If an insurer sees many unrelated claimants all being treated by the same chiropractor and represented by the same lawyer after accidents, it may indicate a coordinated fraud operation.

- **Delayed Injury Report:** The claimant adds an injury claim days or weeks after the accident, especially if the property damage claim was nearly closed. *Rationale:* Sometimes fraudsters see an opportunity and later claim injuries ("by the way, my neck started hurting") to reopen a payout. Genuine accident injuries are usually apparent and documented promptly. A **late assertion of injury** raises suspicion that it's an afterthought to extract more money.

- **False Vehicle Theft or Damage Claims:** Reporting a vehicle as stolen or vandalized when the owner actually disposed of it, or damaging one's own car intentionally (for example, to get an insurance payout on a failing older vehicle).

- **No Evidence of Forced Entry or Hotwiring:** In a theft claim, the recovered vehicle has no signs of break-in, or the owner still has all keys. *Rationale:* A true theft of a locked car typically leaves broken glass or forced locks. Lack of such evidence suggests the "theft" might be staged by the owner (e.g., they might have given the key to an accomplice). Insurers also check if an anti-theft device was installed; if yes and the car still vanished without a trace, it's suspicious.
- **Vehicle Recovered Burned/Stripped with Owner Unreachable:** The car turns up destroyed (burnt or parts missing) and the owner oddly cannot be reached immediately. *Rationale:* Fraudulent owners often burn or dump the car to eliminate evidence and then claim theft. If the owner is strangely indifferent or *too* eager to settle, it could be a sign they orchestrated the loss. Arson to one's vehicle is a subset of hard fraud.
- **Financial Stress or Vehicle Upside-Down:** The owner had recent money problems or the car loan balance far exceeded the car's value. *Rationale:* Motive is a factor. If records show the insured was in

financial distress or owing more than the car is worth ("upside-down" on loan), a sudden theft claim is a red flag. People in such situations have an incentive to stage a theft or destruction to get insurance money (a tactic sometimes called **vehicle dumping** [17]). Insurers may quietly investigate the owner's finances or whether the car was put up for sale pre-incident.

- **Previous Claims History:** The insured has made multiple theft or vandalism claims in the past, or on multiple vehicles. *Rationale:* A **prior loss history** of similar claims significantly raises suspicion [18]. Repeat "bad luck" with stolen or burned cars suggests a pattern of intentional losses. Insurers share data (through NICB databases, etc.) to spot if the same person or VIN has numerous payouts.

- **Claiming Pre-Existing Damage:** Including old vehicle damage in a new claim (or claiming a recent incident caused it).

- **Mismatched Damage Aging:** The damage claimed shows rust, wear, or paint mismatch indicating it existed before the reported accident. *Rationale:* Investigators assess if dents/scratches are fresh. **Patterns or wear on damage** (e.g., rust in a supposed new scratch, uniform weathering) show it wasn't from the recent event [19]. If a claimant tries to bundle old damage repair into a new accident claim, forensic inspection often reveals it.
- **Quick Policy Uptick Before Claim:** The owner added comprehensive/collision coverage or increased limits shortly before the claim for that damage. *Rationale:* If someone suddenly insures an old dent or an already cracked windshield by upping coverage and then, as if on cue, files a claim on it, it's likely pre-existing. Insurers monitor changes in coverage: **recent coverage changes followed by a claim** are a known fraud indicator [20].
- **Inconsistent Damage Location:** The location of certain damages doesn't align with the accident specifics. *Rationale:* For example, a claim from a rear-end collision that also includes unrelated side panel damage. If the mechanics of the accident couldn't cause the additional damage, it implies the claimant is supplementing the real damage with pre-existing issues. Accident reconstruction experts or adjusters will flag such inconsistencies.

Property Insurance Fraud

- **Inflated Repair/Construction Costs:** Exaggerating the cost of repairing property damage (house, building) after an event, often in collusion with contractors or public adjusters.
- **Outlier High Contractor Estimate:** The repair estimates submitted are significantly higher than normal for the type of damage or than the insurance company's own estimate. *Rationale:* A dishonest contractor may **inflate repair costs or perform unnecessary work** [21] to help the insured profit (and maybe take a cut). If one contractor's quote is exorbitant or includes ambiguous fees, the insurer will compare it to industry benchmarks. Large disparities or use of all high-end materials without necessity can indicate padding.
- **Contractor-Insured Relationship:** The insured uses a specific contractor or public adjuster who approaches them unsolicited (e.g., after a disaster) and pressures for a quick claim. *Rationale:* Fraudulent contractors often canvas neighborhoods after storms, urging owners to claim damage. Red flags include a contractor instructing the homeowner **not to contact the insurer until after repairs** [22], or offering to handle all paperwork (possibly to inflate the claim). If the same contractor or adjuster is involved in many questionable claims, insurers become wary.
- **Charges for Undone Work:** The invoice includes services or repairs that were never actually completed, or materials not actually used. *Rationale:* If an audit or re-inspection finds that some claimed repairs can't be verified (e.g., a wall said to be replaced is the original), that means the bill

was padded. Insurers sometimes hold back some payment until proof of repair – if a contractor demands full payment upfront or refuses post-repair inspection, it's suspicious.

- **Early Settlement Pressure:** The insured or their contractor pushes for a fast, lump-sum settlement before providing usual documentation (like detailed inventory of damages or repair invoices). *Rationale:* Insisting on immediate payment and resisting scrutiny is a red flag. Legitimate claims usually proceed through inspections and documentation. Fraudsters try to get cash in hand quickly, before inconsistencies are noticed. An insured **applying pressure for early settlement** or only wanting to communicate via certain channels can signal a setup.

- **Arson or Deliberate Damage:** Intentionally destroying property (often by fire) to collect insurance money [23]. Fire is common since it can obliterate evidence of intentional acts.

- **Fire Origin and Spread Are Suspicious:** Investigators find signs of accelerants (like gasoline) or multiple separate fire start points. *Rationale:* **Arson** frauds often leave forensic clues – unusual burn patterns, empty fuel canisters, or removed valuables. If the fire marshal report indicates the fire was incendiary or of "undetermined origin" with hints of accelerant, the claim will be investigated as potential fraud.
- **Property Contents Removed Prior:** Little to no furniture or personal items were present during the loss (or high-value items were reportedly "in storage" offsite). *Rationale:* If a homeowner's prized belongings fortuitously weren't around during a destructive fire or burglary, it suggests foreknowledge. For example, if just weeks before a fire the owner moved jewelry and important documents elsewhere, that's suspicious timing.
- **High Insurance Recently Obtained:** The coverage on the property was increased shortly before the incident, or previously uninsured items were added. *Rationale:* A classic sign: an owner boosts their home insurance or schedules a bunch of valuables on the policy, then a fire or theft occurs. This **recent coverage change** (especially if within days/weeks) is a red flag [20] because it may indicate planning. Insurers often look back at policy change history when a large first-time claim is filed soon after.

- **Owner's Whereabouts and Alibi:** The incident occurs when the owner is conveniently away (e.g. "on vacation") or at an odd time (middle of night when house empty). *Rationale:* While not proof, an *all too perfect* alibi or absence during a fire/theft can be engineered. Investigators will consider if the owner's story checks out. Multiple red flags — like financial troubles combined with being away during the fire and a new insurance increase — together build a strong fraud suspicion case.

- **False or Exaggerated Theft/Burglary:** Claiming that items were stolen which never existed, or overstating the value/quantity of items stolen in a real burglary.

- **Lack of Forced Entry Evidence:** A theft claim with no signs of break-in (doors, windows intact) and alarm systems (if any) not triggered. *Rationale:* If an insured reports a burglary but investigation finds no physical evidence, it might be fabricated. It's possible to have theft without force (e.g., if door left open), but insurers will question a large theft claim under such circumstances. They may verify police reports – if the insured didn't promptly involve law enforcement, that's a big red flag.
- **Unverified High-Value Items:** The claim list includes many expensive items (jewelry, electronics, art) but the insured has no receipts, photos, or appraisals for them. *Rationale:* People committing fraud often **claim items were stolen that never were** or never owned [24]. If a claimant cannot produce any proof of purchase or ownership for big-ticket items they say were lost, skepticism is warranted.

Insurers may request proof or see if the items even existed (sometimes they ask for the model/serial numbers, etc.).

- **Entire Inventory Lost is High-End:** The loss inventory seems "too perfect," e.g., only the most expensive and newest items were stolen or destroyed, and all of them. *Rationale:* In real thefts, criminals grab what's accessible; it's rare that *only* the insured's priciest jewelry and newest electronics are gone without a trace. A claim that conveniently totals to a very high amount, just under policy limits, with little variation (and little in the way of lower-value items) suggests it was curated by the insured to maximize payout. Insurers use databases of typical household contents and values – a claim far outside the statistical norm (like five big-screen TVs from a one-bedroom apartment) raises alarms.

- **Delayed or Police-Unreported Claim:** The insured delays reporting the alleged theft to the insurer or never filed a police report. *Rationale:* If weeks pass before the claim or the **loss was never reported to authorities**, it's highly suspect. Genuine victims usually report crimes immediately. A delay might mean the insured needed time to concoct the story or "dispose" of items. Insurance policies also often require timely notice; a late claim can be grounds for denial if fraud is suspected.

- **Misrepresentation of Pre-Existing Damage or Wear:** Claiming long-term issues as sudden damage from a covered peril.

- **Old Damage Presented as New:** The insured claims a recent storm or accident caused damage that appears aged – e.g., old water stains or structural issues now attributed to a fresh incident. *Rationale:* If an inspector finds that supposed storm damage actually shows signs of long-term neglect or prior repair patches, it wasn't caused by the event. For example, claiming an **old leaky roof** was the result of last week's hail. Investigators will differentiate new hail dents from old ones (looking at oxidation, patina, etc.) [25] . Pre-existing issues are not covered, so fraudsters lie about the timing.
- **Maintenance Neglect Framed as Accident:** Damage that really came from poor maintenance or wear-and-tear is blamed on a sudden event. *Rationale:* Home insurance doesn't cover gradual damage (rot, mold, deterioration). If an insurer sees, for instance, a rotted beam after a minor earthquake and the claim is for "earthquake damage," they'll suspect the earthquake is just being used as an excuse. Any claim where the **damage is inconsistent with the reported cause or timing** (e.g., extensive long-term termite damage claimed as one-day hurricane damage) will trigger a closer look.
- **Multiple Unrelated Claims Together:** The insured bundles several damage claims from supposedly one incident that don't logically all stem from that single cause. *Rationale:* A fraud-minded homeowner might use one real event to get other unrelated fixes done on the insurance's dime ("one claim to rule them all"). For instance, after a small kitchen fire, they also claim basement water damage and an older roof leak as part of the same incident. If the scope of loss seems artificially expanded, it's likely they are tacking on pre-existing problems to the legitimate claim.

Cross-Cutting Red Flags: Across all insurance types, certain red flags apply generally. For example, a **history of frequent claims** or similar past questionable claims by the same person is a strong indicator of opportunistic fraud [18] . Also, **claims made soon after policy inception or shortly before policy expiration** (possibly hoping insurer's vigilance is lower) are suspect [26] [27] . **Inconsistent accounts or changing stories**, lack of witnesses, and reluctant cooperation from the claimant are universal warning signs. Insurers leverage these factors in rule-based fraud screens regardless of claim type.

Data Model for Fraud Detection (Dimensions & Facts)

To detect the above fraud factors, a comprehensive **data model** is designed for each insurance domain (Health, Auto, Property). The model uses a **dimension and fact table** approach (star schema) – capturing granular claim details and reference data – aligning with industry standards for insurance data warehouses [28]. Each table below lists its key columns, data types, and descriptions. This schema supports both rule-based flagging (via explicit indicator fields and relationships) and machine learning feature extraction by storing rich, historical data.

Health Insurance Data Model

The health insurance model tracks patients, providers, policies, and detailed claims (including visits, treatments, labs, medications, and payments). It is structured to identify patterns like upcoding (via procedure codes and costs), phantom billing (via unlinked services), and other medical fraud indicators.

Patient Dimension (Health)

Column Name	Data Type	Description
patient_id	String (UUID)	Unique patient identifier (primary key).
full_name	String	Patient's name.
date_of_birth	Date	DOB (helps detect identity fraud and age-related norms).
gender	String	Gender of the patient.
address	String	Home address (for geographic analysis of claims).
join_date	Date	Date the patient was added to the insurer's system or policy.
policy_id	String (UUID)	Foreign key to Policy Dimension, indicating current coverage.
medical_history_flags	String	Optional summary flags (e.g., chronic conditions) – for context in claims (e.g., multiple expensive treatments without relevant history can flag fraud).

Rationale: The Patient dimension stores demographic info that can be used to detect anomalies (e.g., multiple patients sharing an address making similar claims might indicate an identity/medical fraud ring). Linking to **Policy** allows checking coverage specifics and enrollment timing (e.g., new patient with immediate large claims). Past medical history context helps in fraud rules (for example, a claim for a condition the patient doesn't have on record could be bogus).

Provider Dimension (Health)

Column Name	Data Type	Description
provider_id	String (UUID)	Unique identifier for a healthcare provider (doctor, clinic, hospital).
name	String	Provider or facility name.
provider_type	String	Type of provider (e.g., Individual Physician, Clinic, Hospital, Lab).
specialty	String	Medical specialty (if applicable, e.g., Orthopedics, General Practice).
license_number	String	Professional license ID (for validation and detecting bogus providers).
hospital_affiliation	String	Affiliated hospital/clinic name (if any).
region	String	Geographic region or address for the provider.
fraud_score	Float	*Optional:* A risk score from prior audits (higher if past fraud issues).

Rationale: The Provider dimension captures information to identify **provider-specific fraud patterns**. For example, analyzing claims by provider can reveal if a provider has an unusually high volume of expensive procedures (upcoding) or uncommon specialties performing unrelated services. The **license_number** helps confirm legitimacy (and catch "impersonating a professional" fraud [8]). **Provider_type and specialty** help verify that billed procedures make sense (e.g., a podiatrist billing for heart surgery would be a glaring red flag). The optional fraud_score or flags can incorporate external data (e.g., if NICB or a health bureau has flagged the provider before).

Policy Dimension (Health)

Column Name	Data Type	Description
policy_id	String (UUID)	Policy identifier (primary key).
plan_type	String	Type of insurance plan (e.g., HMO, PPO, Medicare, etc.).
coverage_start_date	Date	Policy effective start date.
coverage_end_date	Date	Policy end date (if applicable or renewal period).
coverage_limits	String	Key coverage limits (e.g., max claim amount, specific caps like therapy sessions limit).
policyholder_id	String (UUID)	ID of the primary insured (could link to Patient or a separate Policyholder dimension if group policy).

Column Name	Data Type	Description
premium_amount	Float	Premium paid (can hint at policy value; not directly fraud but context).
fraud_flags	String	*Optional:* Indicators like “reinstated policy”, “recently upgraded coverage” – useful for fraud context (e.g., claim soon after upgrade).

Rationale: The Policy dimension provides context such as when coverage began (critical for rules like “recent policy, quick claim” [26]) and coverage limits (to see if claims push these limits frequently). **Plan type** can be relevant (certain fraud schemes target specific programs like Medicare). Knowing coverage limits and terms allows detection of unusual patterns (e.g., multiple high-value claims just under limits). If policy has just been renewed or coverage expanded, a large claim immediately could be flagged.

Claim Fact (Health)

Column Name	Data Type	Description
claim_id	String (UUID)	Unique claim identifier (primary key).
patient_id	String (UUID)	Foreign key to Patient Dimension (who is the claimant/patient).
policy_id	String (UUID)	Foreign key to Policy Dimension (which policy covers this claim).
provider_id	String (UUID)	Foreign key to Provider Dimension (who submitted/provided service).
claim_date	Date	Date the claim was filed.
service_date	Date	Date of service (or start date if a range) that the claim covers.
claim_type	String	Type of claim (e.g., Inpatient, Outpatient, Pharmacy, Dental).
diagnosis_code	String	Primary diagnosis code (ICD) for the claim (if applicable).
total_claim_amount	Float	Total amount billed on the claim.
approved_amount	Float	Amount approved/paid by insurer (after adjustments).
claim_status	String	Status (e.g., Pending, Approved, Denied, Under Investigation).
suspicious_flags	String	*Optional:* Flags set by rules engine (e.g., “High$ Upcoding Risk”, “Duplicate Claim Suspected”).

Rationale: The Claim fact is the central fact table summarizing each insurance claim. It links to **Patient**, **Provider**, and **Policy** to allow multi-dimensional analysis (e.g., all claims by a provider, or all claims under a certain policy type). Key fields enable fraud detection:

- Comparing **claim_date** vs **coverage_start_date** (from Policy) reveals if a claim came soon after policy inception.
- **Service_date** vs claim_date gap can show delays in filing (late reporting is suspicious in some cases).
- **Diagnosis_code** along with procedures in sub-tables helps catch mismatches (e.g., billing for treatments that don't fit the diagnosis).
- **total_claim_amount** allows rules like "claims just below a threshold" or outlier detection (an unusually high claim for one patient).
- If **claim_status** is "Under Investigation" or if **suspicious_flags** are set, those come from rule-based triggers. This table can feed a fraud case management system.

Each claim may have multiple associated service details (visits, treatments, labs, etc.), which are broken out into separate fact tables (below) for granular analysis.

Doctor Visit Fact (Health)

Column Name	Data Type	Description
visit_id	String (UUID)	Unique ID for a patient's visit/encounter (primary key).
claim_id	String (UUID)	Foreign key to Claim Fact (if this visit resulted in a claim).
patient_id	String (UUID)	Foreign key to Patient Dimension (who visited).
provider_id	String (UUID)	Foreign key to Provider Dimension (who was visited).
visit_date	DateTime	Date and time of the visit/appointment.
visit_type	String	Type of visit (e.g., Office, Emergency, Telehealth).
chief_complaint	String	Brief reason for visit (e.g., "back pain").
notes_flag	Boolean	Indicator if detailed clinical notes are available (for auditing).

Rationale: Each **Doctor Visit** record represents an encounter between patient and provider. Fraud detection uses visit data to verify claims: a **phantom billing** scenario can be flagged if a claim exists without a corresponding visit record. For example, if a claim's service_date doesn't match any logged visit for that patient, it's suspicious. Also, analyzing **visit_type** and timing can reveal excessive frequency of visits (patient or provider over-utilization). The **notes_flag** indicates whether supporting documentation exists; phantom or upcoded services often lack proper notes. By linking visits to claims, we ensure every billed service has an actual visit. Visits with no claims (or vice versa) can both be insightful (some fraudsters might bill without actual visits, or have patients come in for nothing significant but bill heavily).

Treatment/Procedure Fact (Health)

Column Name	Data Type	Description
treatment_id	String (UUID)	Unique ID for a treatment/procedure entry (primary key).

Column Name	Data Type	Description
claim_id	String (UUID)	Foreign key to Claim Fact (the claim this procedure is part of).
provider_id	String (UUID)	Foreign key to Provider Dimension (who performed/billed it).
procedure_code	String	Procedure code (CPT/HCPCS code for the service provided).
procedure_desc	String	Description of the procedure (from code master, e.g., "MRI Scan – Brain").
quantity	Integer	Quantity of this service (if applicable, e.g., number of sessions).
unit_cost	Float	Billed cost per unit (fee for one service unit).
total_cost	Float	Total cost = unit_cost * quantity (for this line item).
diagnosis_code	String	Diagnosis code linked to this procedure (if specific to line item).
is_unbundled_flag	Boolean	*Optional:* Indicator if this procedure was part of a bundle that was split (set via rules).

Rationale: The Treatment fact table lists individual billed services/procedures on claims. Having each **procedure_code** with cost allows detection of **upcoding** (e.g., if a usually inexpensive procedure appears with a code for a more expensive one, or unusually high unit_cost). By summing or analyzing **total_cost** across procedures, we can flag if a provider bills many high-cost procedures. The **diagnosis_code** at line-item level allows checking that each procedure is medically justified (e.g., physical therapy procedure linked to a back-pain diagnosis, etc.). If a procedure is performed in a suspicious combination (like multiple separate codes that normally bundle), a rule can set **is_unbundled_flag** to true. This table also helps find **double billing**: if two treatment entries exist with the same procedure_code, date, and patient but from different claims (or same claim twice), that's a duplicate charge indicator. Because each procedure is a separate record, analytics can find patterns like a provider performing an unusually high number of a certain expensive procedure code compared to peers [6] .

Laboratory Result Fact (Health)

Column Name	Data Type	Description
lab_id	String (UUID)	Unique ID for a lab test entry (primary key).
claim_id	String (UUID)	Foreign key to Claim Fact (claim covering this test).
patient_id	String (UUID)	Foreign key to Patient Dimension (who the test is for).
lab_test_code	String	Code for the lab test (e.g., LOINC code or internal test ID).
lab_test_name	String	Name/description of test (e.g., "Complete Blood Count").

Column Name	Data Type	Description
result_value	String	Result value (or status, e.g., numeric result or “Positive/ Negative”).
test_date	Date	Date the lab test was conducted.
cost	Float	Cost billed for the lab test.
ordering_provider_id	String (UUID)	Provider who ordered the test (if different from billing provider).

Rationale: The Lab fact table records individual lab tests billed. **Phantom billing** often involves fake lab tests, so verifying that each lab test corresponds to an actual result (non-null **result_value**) and that the patient’s medical record indicates it was needed is key. Unusual patterns such as very expensive lab panels ordered for routine checkups can be spotted here. The **lab_test_code/name** helps identify if the test is appropriate for the diagnosis (e.g., cancer markers billed for a simple injury would be odd). Also, if a provider is billing a high volume of certain expensive tests (like genetic tests) across many patients, we can catch that trend by aggregating this table. **Cost** helps flag overpriced tests or duplication. By linking **ordering_provider_id**, we can detect if one doctor’s name is used to justify many tests possibly without their knowledge (identity misuse). This granular lab data supports rules like “test ordered without corresponding diagnosis” or ML features like number of tests per visit.

Medication/Pharmacy Fact (Health)

Column Name	Data Type	Description
med_id	String (UUID)	Unique ID for a medication dispensed or claimed (primary key).
claim_id	String (UUID)	Foreign key to Claim Fact (the claim for this prescription).
patient_id	String (UUID)	Foreign key to Patient Dimension.
drug_code	String	Drug identifier (e.g., NDC code or drug name).
drug_name	String	Name of medication (brand/generic).
quantity	Float	Quantity dispensed (e.g., number of pills or volume).
days_supply	Integer	Days of supply for the medication (if applicable).
prescribing_provider_id	String (UUID)	Provider who prescribed the drug.
pharmacy_id	String	Identifier of pharmacy (if relevant, or could use Provider Dim for pharmacies).
cost	Float	Cost billed for the medication (total).

Column Name	Data Type	Description
is_refill_flag	Boolean	Indicates if this entry is a refill.

Rationale: Medication claims are a frequent area of fraud (e.g., **doctor shopping** for controlled substances, billing brand-name drugs but dispensing generics, etc.). This table allows tracking every prescription claim. By analyzing **quantity** and **days_supply**, one can flag if a patient is getting excessive medication (e.g., overlapping refills from multiple doctors – potential diversion or abuse). **Prescribing_provider_id** lets us see if one doctor is writing an inordinate number of scripts (possible pill mill). The **drug_code/name** and **cost** help catch **upcoding in pharmacy** (e.g., billing for a more expensive drug than given) or **phantom pharmacy billing** (medications billed with no record of pickup). If a medication is expensive or controlled, rules may ensure diagnoses support its use. Cross-checking between medication and treatment tables can reveal inconsistencies (like expensive medications without any office visits or lab tests in the record). This data also supports ML features such as "number of opioid prescriptions for patient in last 6 months" to detect abuse or fraud.

Payment Fact (Health)

Column Name	Data Type	Description
payment_id	String (UUID)	Unique payment transaction ID (primary key).
claim_id	String (UUID)	Foreign key to Claim Fact (the claim this payment is for).
payee_type	String	Who was paid (e.g., Provider, Patient, Pharmacy).
payee_id	String	Identifier of payee (e.g., provider_id if paid to provider, patient_id if reimbursing patient).
payment_date	Date	Date of payment.
paid_amount	Float	Amount paid in this transaction.
payment_method	String	Method (e.g., Check, EFT, Credit).
is_final_payment	Boolean	Whether this is final/full payment for the claim (vs. partial or adjusted payment).

Rationale: The Payment fact table logs each payout related to a claim. **Individual payment records** are part of industry-standard data models [29] to track how claims are disbursed. For fraud detection, this provides an audit trail: if a claim was paid multiple times (duplicate payments) or overpaid, it would show here. It also identifies **who received the money** – useful in schemes where, say, a patient and provider might split fraudulent proceeds. Unusual patterns, such as payments issued to entities unrelated to the provider (potential shell companies) or multiple checks to different parties for one claim, can be flagged. This also helps in recovery/subrogation efforts if fraud is confirmed. From a ML perspective, features like "percentage of claimed amount paid" could indicate something (fraudulent claims might be caught and only partially paid or denied). Rapid, high-frequency payouts to the same provider can also raise red flags.

Auto Insurance Data Model

The auto insurance model captures details of vehicles, policyholders, accidents (incidents), claim details (for vehicle damage and injuries), repair estimates, evidence, and payments. This structure enables detection of patterns like staged accidents (via repeat participants or odd incident details), inflated repair costs (via detailed estimate line items), and exaggerated injury claims (via injury details versus vehicle damage).

Insured Person Dimension (Auto)

Column Name	Data Type	Description
insured_id	String (UUID)	Unique identifier for an insured individual (primary key).
name	String	Name of the person (policyholder or claimant).
date_of_birth	Date	Date of birth (for age – young drivers, etc., could be relevant).
license_number	String	Driver's license number (to identify unique drivers).
address	String	Address of the insured (useful for detecting geographic clusters or suspicious ring).
policy_id	String (UUID)	Foreign key to Auto Policy Dimension (if this person is a policyholder on a policy).
customer_since	Date	Date the person became a customer (policy start for them).
prior_claim_count	Integer	*Optional:* Number of prior claims filed by this person (historical).

Rationale: The Insured dimension holds information about individuals involved in auto policies or claims. It helps catch fraud like **serial claimants** – e.g., if the same person has many prior claims, the **prior_claim_count** can trigger extra scrutiny [18]. The **address** can expose fraud rings (multiple "unrelated" claimants using the same address or small area for staged accidents). **License_number** uniquely identifies drivers across policies (someone might be listed on multiple policies under aliases). Age and customer_since are useful for context; for instance, an insured who joined last week and immediately had a big accident stands out. This dimension can be joined to **Accident/Claim** records to see if the same person appears as driver or claimant frequently (pattern of staged accidents).

Vehicle Dimension (Auto)

Column Name	Data Type	Description
vehicle_id	String (UUID)	Unique vehicle identifier (primary key).
VIN	String	Vehicle Identification Number (unique serial for the vehicle).
make	String	Vehicle make (manufacturer), e.g., Toyota.
model	String	Vehicle model, e.g., Camry.

Column Name	Data Type	Description
year	Integer	Model year of the vehicle.
owner_insured_id	String (UUID)	Foreign key to Insured Person (who is the owner).
current_value	Float	Approximate current market value of vehicle (from underwriting data).
vehicle_type	String	Type/class (Sedan, SUV, Truck, etc.).
anti_theft_installed	Boolean	Whether an anti-theft device is installed (for theft claim context).

Rationale: The Vehicle dimension centralizes car information, enabling fraud checks like: if a claimed damage or theft doesn't match the **vehicle_type** or value. For example, claiming very high repair costs on an old car valued low is suspicious if it approaches total value. **VIN** helps cross-reference the vehicle's history (was it previously salvaged? previously claimed as stolen?). Patterns such as the same VIN appearing in multiple claims over time (especially total losses) could indicate fraud. Knowing the **owner_insured_id** links the vehicle to a person, so we can see if one person is associated with many vehicles with claims (possible fraud ring or **vehicle dumping** scenario if they cycle through cars). The **anti_theft_installed** flag is useful: if a vehicle with an advanced anti-theft system is reported stolen without force, that's a red flag as noted. Also, make/model/year let adjusters see if claimed repair costs make sense; e.g., luxury car parts vs. economy car.

Auto Policy Dimension

Column Name	Data Type	Description
policy_id	String (UUID)	Policy identifier (primary key) for auto insurance policy.
policyholder_id	String (UUID)	Insured Person who is the primary policyholder (FK to Insured Dim).
policy_start_date	Date	Policy effective start date.
policy_end_date	Date	Policy expiration/cancellation date.
coverage_types	String	Coverage included (e.g., Liability, Collision, Comprehensive, Personal Injury Protection).
coverage_limits	String	Key limits (e.g., liability limit per accident, collision deductible, etc.).
vehicle_ids	String	List of vehicles covered (or could be a separate link table; simplified here as comma-separated or count of vehicles).
premium_annual	Float	Annual premium amount.
last_claim_date	Date	Date of last claim (if any prior, for quick reference).

Column Name	Data Type	Description
flags	String	*Optional:* Notes like “New Policy”, “Prior Lapses in coverage” etc. indicating risk factors.

Rationale: The Auto Policy dimension provides context to evaluate claims: a claim’s timing relative to **policy_start_date** is crucial (new policy with quick large claim = suspect). **Coverage_types** and limits let us know what can be claimed – e.g., if a claim includes a type not covered, that’s fraudulent (or if multiple policies overlap for same incident, potential fraud). The list of covered **vehicle_ids** ties policies to vehicles. The **policyholder_id** connects to the person; if one person has multiple active policies (uncommon) or frequently changing insurers, that’s interesting. **Last_claim_date** can highlight if multiple claims are occurring in short succession. The policy flags might include whether the policy was reinstated recently or upgraded coverage, which, as mentioned, often precedes fraudulent claims. This dimension basically helps isolate whether a claim falls under any suspicious policy scenarios (new policy, high coverage recently chosen, etc.).

Accident/Incident Fact (Auto)

Column Name	Data Type	Description
incident_id	String (UUID)	Unique ID for an accident/incident (primary key).
claim_id	String (UUID)	Foreign key to Claim Fact (the claim resulting from this incident).
incident_date	DateTime	Date and time the accident or loss occurred.
location	String	Location of incident (address or coordinates).
incident_type	String	Type of incident (Collision, Theft, Fire, Vandalism, etc.).
police_report_number	String	Police report ID (if reported to police).
weather_conditions	String	Weather/road conditions (if applicable, for context).
incident_description	String	Text description of what happened (from claims report).
fault_indicator	String	Indication of fault (e.g., Insured at-fault, Third-party at-fault, Undetermined).
witness_present	Boolean	Whether independent witnesses were present (according to report).

Rationale: The Incident fact table details the context of the claim event. It helps detect staged or suspicious accidents by analyzing the **incident_description** and metadata:
- **incident_date/time**: Patterns like accidents always at off-hours or specific days could hint at planned events. Also, ties into verifying delays (if claim filed long after incident_date, why?).
- **location**: If many accidents happen at the same spot or neighborhood involving the same people, that’s a red flag (could indicate a known staging location).
- **incident_type**: Combined with claim details, ensures consistency (e.g., if incident_type is Theft but the

claim has collision damage repairs, something's wrong).
- **police_report_number**: Lack of a police report for a serious accident is a red flag. Or multiple claims sharing one police report number (potential organized fraud).
- **fault_indicator**: Staged accidents often try to pin fault on the innocent party. If our insured is repeatedly "not at fault" in dubious accidents, they might actually be orchestrating being the victim. Conversely, if always at-fault but with similar patterns, they could be perpetrator.
- **witness_present**: Staged accidents often have either fake witnesses or no independent witnesses. A flag for no witnesses could combine with other factors (like multiple passengers who are claimants) to suggest fraud.

This table is essential for rule-based detection like: *"Multiple accidents with same participants or same location"*, *"Incident occurred before policy start"* (immediate red flag if someone backdates an incident). It also feeds ML models with features about how accidents happen.

Claim Fact (Auto)

Column Name	Data Type	Description
claim_id	String (UUID)	Unique auto claim identifier (primary key).
policy_id	String (UUID)	Foreign key to Auto Policy (which policy covers this claim).
insured_id	String (UUID)	Foreign key to Insured Person (the person filing the claim or insured driver).
vehicle_id	String (UUID)	Foreign key to Vehicle Dimension (vehicle involved in claim).
incident_id	String (UUID)	Foreign key to Incident Fact (details of the accident/event).
claim_date	Date	Date the claim was reported to the insurer.
claim_type	String	Claim category (e.g., "Collision Damage", "Theft", "Injury/ Bodily Injury", "Comprehensive (fire/flood)").
claimed_damage_amount	Float	Total amount of property damage claimed (vehicle repair or value).
claimed_injury_amount	Float	Total amount claimed for injuries (medical bills, etc. if any).
num_injuries_claimed	Integer	Number of individuals claiming injury in this incident.
claim_status	String	Status (Open, Closed, Denied, Under Investigation, etc.).
suspicious_flags	String	*Optional:* Flags from business rules (e.g., "High Repair Cost", "Repeat Claimant").

Rationale: The auto Claim fact ties together key pieces: which policy, person, vehicle, and incident are involved. This allows multifaceted analysis: for instance, joining Vehicle and Insured can show if the same

person had multiple claims on different vehicles (potentially buying cheap cars and “totaling” them repeatedly). Important fields:
- **claim_date vs incident_date**: A long delay in reporting (incident happened much earlier than claim_date) is suspicious – could be a sign of fabricated or late circumstances.
- **claim_type**: Combined with incident_type to ensure consistency (if claim_type is Collision but incident_type is Theft, data discrepancy). Also, if one claim involves multiple types (e.g., both theft and collision), it might actually be two events being merged improperly.
- **claimed_damage_amount vs vehicle_value**: If someone claims repairs or total loss far exceeding the car’s market value, it may indicate inflation or fraud (or it becomes a total loss scenario). Many fraudulent claims aim just below total loss threshold to get repair money that exceeds actual costs.
- **claimed_injury_amount & num_injuries_claimed**: Puts numbers to injury claims. For example, if **num_injuries_claimed** is high (lots of people in car) and the accident was minor, that’s a red flag. Patterns like always having the maximum number of passengers claiming injury in every accident a person has is suspicious. Also, a large claimed_injury_amount with minimal vehicle damage is a key discrepancy (soft fraud).
- **suspicious_flags**: The system might auto-flag things like “accident soon after policy start”, “vehicle not inspected” etc., captured here for investigators.

This Claim fact is the hub that links to more detailed breakdowns (repair estimates, injuries, payments) below, enabling both summary analytics and drill-down.

Repair Estimate Fact (Auto)

Column Name	Data Type	Description
estimate_id	String (UUID)	Unique ID for a repair estimate line item (primary key).
claim_id	String (UUID)	Foreign key to Claim Fact (which claim this estimate is for).
vehicle_part	String	Name/Code of vehicle part or repair category (e.g., “Front Bumper”, “Paint Labor”).
damage_description	String	Description of damage to that part (e.g., “cracked, needs replacement”).
estimated_cost	Float	Estimated cost for this part repair/replacement (from the shop).
is_oem_part	Boolean	Whether the estimate is using OEM (original) parts (often costlier) or aftermarket.
repair_shop_id	String (UUID)	Identifier for repair shop/garage providing the estimate (could link to a Shop/Provider dim).
labor_hours	Float	Estimated labor hours for this line (if applicable).
final_cost	Float	Final approved cost for this item (after adjuster review, if different).

Rationale: The Repair Estimate table itemizes the **vehicle damage repair costs**. This granularity is crucial to detect **inflated repair bills** [10] and possibly phantom repairs:
- By summing **estimated_cost** across items and comparing with typical repair databases or the car's value, outliers can be flagged. For example, a $10,000 total estimate for minor fender damage would stand out.
- **vehicle_part** and **damage_description** allow investigators or algorithms to check consistency with the accident. If a claim includes parts that shouldn't be affected (e.g., claiming damage to the rear when the accident was front-end) – that's evidence of fraud.
- **is_oem_part**: Fraudulent shops might quote only expensive OEM parts even when not needed, to inflate cost. A rule might flag if all parts are OEM for an old car where usually aftermarket would suffice.
- **repair_shop_id**: We can see if the same shop is associated with many high estimates or claims (potential collusion). If a specific garage appears frequently in questionable claims, that's a link.
- **labor_hours**: Extremely high labor times for simple repairs could indicate padding. For instance, 50 hours of labor to replace a bumper is unrealistic.
- **final_cost** vs **estimated_cost**: If adjusters significantly cut costs, that suggests the initial was inflated. Patterns where a particular estimator or shop always overshoots (and gets cut) can be tracked.

By having each part's estimate, ML models can even learn what typical damage costs are given accident conditions, and flag anomalies. Rule-based checks like "sum of one category (like paint) exceeds threshold" can also be applied.

Injury Detail Fact (Auto)

Column Name	Data Type	Description
injury_id	String (UUID)	Unique ID for an injury claim entry (primary key).
claim_id	String (UUID)	Foreign key to Claim Fact (the auto claim this injury is associated with).
person_name	String	Name of injured person (could be the insured or a third-party).
person_role	String	Role of person (Driver, Passenger, Third-Party, Pedestrian, etc.).
injury_description	String	Description of injury (e.g., "whiplash", "broken arm").
medical_provider	String	Name or ID of medical provider/hospital treating the injury (optional link to Provider Dim if integrated with health data).
claimed_medical_cost	Float	Claimed medical cost for this person's injuries (if itemized).
injury_severity	String	Reported severity (Minor, Moderate, Severe – possibly subjective).
treatment_duration	Integer	Treatment duration in days (or number of sessions, if known).

Rationale: The Injury Detail table lists each individual injury claim resulting from an auto accident. This is critical for detecting **exaggerated injury fraud**:
- **person_role** and number of entries help verify if the count of injured makes sense (e.g., 5 passengers in a 2-seater car would be fraudulent).
- **injury_description** allows comparison with the accident: if someone claims "serious back injury" from a

minor fender-bender, that's suspect. Also, if all claimants from one accident have identical vague injuries like "soft tissue neck pain", it might be coached fraud.
- **claimed_medical_cost**: Summing these should equal the **claimed_injury_amount** in the Claim fact. If extremely high relative to the accident or relative to typical treatment of that injury, it's a red flag. For example, \$100k for whiplash treatment is excessive.
- **medical_provider**: If all injured claimants coincidentally go to the same clinic or doctor (especially a known high-claims chiropractor), that suggests a fraud ring. Linking to the Provider dimension (if available from health data or via name matching) can expose that pattern.
- **injury_severity** vs actual evidence: If severity is claimed "Severe" but the medical reports (if reviewed) don't substantiate it, or if the treatment_duration is short despite "Severe" labeled, there's inconsistency. Conversely, very long treatment_duration for a "Minor" injury is also odd and could indicate padding.
- This table also helps in looking at **who** is injured in repeated incidents: if the same person appears injured in multiple claims (especially as a passenger in others' cars repeatedly), they might be a professional claimant. By storing names/IDs, we can cross-reference identities of frequent claimants across different claims (common in staged accident rings where the same "victim" pops up).

In summary, Injury details give a more fine-grained view to apply rules like "injury claims inconsistent with vehicle damage" or "repeated injury claims by same person or treated by same provider." These are key to catching fraud that pure vehicle repair data might miss.

Photo Evidence Fact (Auto)

Column Name	Data Type	Description
photo_id	String (UUID)	Unique ID for a photo or video evidence file (primary key).
claim_id	String (UUID)	Foreign key to Claim Fact (the claim this evidence is for).
photo_type	String	Type of media (e.g., Vehicle Damage Photo, Scene Photo, Police Photo, Video).
taken_date	DateTime	Date/time the photo was taken (if metadata available).
description	String	Description of what the image shows (e.g., "front bumper damage").
file_path	String	Storage reference or URL to the image (location in system).
verified_flag	Boolean	Flag if the insurance adjuster verified the photo's authenticity (e.g., matches scene, not stock photo).

Rationale: Photo evidence is often provided to document auto claims. This table doesn't lend itself to numeric analysis, but it's crucial for investigators and possibly for computer vision models in fraud detection (beyond the scope here). However, even in structured analysis:
- The presence or absence of photos (and their timing) can be telling. For instance, if a severe accident claim has **no photos** provided at all, that's unusual nowadays and might be a red flag (why wouldn't the insured provide any evidence?). Conversely, a claim with suspicious photos (maybe reused or inconsistent vehicles) could be caught by an adjuster – the **verified_flag** would be false until checked.

- **taken_date** vs incident_date: If metadata shows photos were taken *before* the incident or long after, that's suspicious (e.g., someone using old damage photos).
- Many staged claims might avoid clear photos or provide only close-ups to hide context. On the other hand, if the same photo shows up on multiple claims (hash or file_path matching), that's outright fraud – a person reusing damage photos for multiple incidents. Storing file references allows such cross-checking.
- Though not a traditional "fact" for quantitative analysis, including Photo evidence in the model ensures a **complete case record**. It supports rule-based workflows (like flagging if photos don't match the vehicle described, or if license plates differ, etc., via human or AI review).

This table primarily supports investigative processes and any advanced image analysis tools. It rounds out the data available for a claim, so nothing is solely kept in unstructured form outside the purview of fraud detection systems.

Payment Fact (Auto)

Column Name	Data Type	Description
payment_id	String (UUID)	Unique ID for a payment transaction (primary key).
claim_id	String (UUID)	Foreign key to Claim Fact (which claim this payment is for).
payee	String	Who was paid (e.g., insured, repair shop, lienholder, third-party claimant).
payee_id	String	Identifier for payee if available (could be insured_id, or shop ID, etc., or blank if one-time payee).
payment_date	Date	Date payment was issued.
amount_paid	Float	Amount of the payment.
payment_purpose	String	Purpose of payment (e.g., "Vehicle Repair Settlement", "Total Loss Payout", "Injury Compensation").
payment_mode	String	Mode of payment (Check, Bank Transfer, etc.).
recovery_indicator	Boolean	Flag if payment is subject to recovery/subrogation (e.g., if fraud or third-party liability is proven later).

Rationale: Similar to the health Payment table, this logs all disbursements for auto claims. It's especially useful in auto because payments might go to various parties (the insured, a body shop, a claimant's lawyer, etc.). Key points for fraud:
- If multiple large payments go to the same **payee** for different claims, and that payee is not obviously a repair chain or legitimate business, it could indicate a straw party or a fraudulent claimant profiting repeatedly. For example, if "John Doe" has received three injury payouts as a third-party claim in different accidents, the data will show that.
- The **recovery_indicator** can be toggled if, say, fraud is suspected after payment and the insurer is trying to recover funds or has not finalized (it's a complex scenario, but tracking that helps in fraud case management).

- We can see if payment was made extremely quickly (payment_date shortly after claim_date) – sometimes insurers rush small claims, but for large ones a too-quick payout might indicate someone internal was bypassing checks (internal fraud possibility or just a note on process).
- **amount_paid vs claimed amounts**: If an insurer negotiated down or only paid partial, that can be because adjusters found the claim excessive. Patterns of consistently paying far less than claimed (because claims were inflated) might highlight a specific customer or provider who always overclaims.
- Also, if an **amount_paid** exactly equals a policy limit, that stands out (potential maxing out as discussed).
- Linking payments to claims and ultimately to people and shops provides the trail of money flow, which is crucial in fraud investigations (following the money to see if, for instance, an insured's relative's shop got paid).

Overall, the Payment fact ensures that every financial transaction is accounted for and can be analyzed for irregularities (like duplicate payments, unusual payees, etc.).

Property Insurance Data Model

The property insurance model covers insured entities, properties, policies, claims, and detailed breakdowns of losses (contractor repair estimates, contents inventories for theft, evidence, and payments). It's designed to catch fraud indicators such as overstated claims (through itemized comparisons), arson/red-flag events (through incident details), and repeat patterns.

Insured Person/Entity Dimension (Property)

Column Name	Data Type	Description
insured_id	String (UUID)	Unique identifier for the insured person or business (primary key).
name	String	Name of the policyholder (individual or company).
address	String	Primary address of insured (if individual) or business address.
type	String	Type of insured (Individual, Business).
contact_info	String	Contact details (phone/email) of insured.
customer_since	Date	Date this customer started with insurer (first policy start).
prior_claim_count	Integer	Number of past property claims by this insured.
fraud_watchlist_flag	Boolean	Flag if insured is on any internal watchlist for suspicious activity.

Rationale: The Insured dimension for property is similar to auto – it identifies the person or entity making claims. **prior_claim_count** is telling; someone with frequent fire or theft claims is high risk for fraud (history of suspicious losses) [18]. If it's a business insured, patterns like repeated burglary claims at the same business could emerge by grouping claims by insured_id. The **fraud_watchlist_flag** might be set if, for example, a previous claim was investigated for fraud even if not proven. Address here is less about accident location (since property location is separate) and more about seeing if multiple "insured" share addresses (possible collusion, though less common than in auto). Still, if multiple distinct insured entities use one

mailing address (like a P.O. box or one apartment known for fraud scams), that's noteworthy. This dimension ties to policies and claims to look for individuals or businesses that stand out.

Property Dimension

Column Name	Data Type	Description
property_id	String (UUID)	Unique identifier for the insured property (primary key).
address	String	Property address (location of the property risk).
property_type	String	Type of property (Single-Family Home, Condo, Commercial Building, etc.).
construction_year	Integer	Year the property was built (for context; older properties might have more pre-existing issues).
construction_type	String	Building structure type (Wood, Masonry, etc. – impacts fire risk).
occupancy	String	Occupancy status (Owner-Occupied, Rented, Vacant, etc.).
home_security_system	Boolean	Whether a security system is present (for theft mitigation context).
appraised_value	Float	Latest appraised value of property (building).
insured_id	String (UUID)	Foreign key to Insured (owner of the property).

Rationale: The Property dimension gives details about the physical asset. Fraud detection uses this to check consistency of claims:
- **address** is key for verifying incidents (e.g., if a hail claim is reported, one can cross-check if hail actually occurred at that location/date). Also, seeing multiple claims from the same neighborhood (especially if same contractor involved) is a red flag for potential fraud rings or serial claims after disasters.
- **property_type** and **occupancy**: If a property was actually vacant but a claim says contents were stolen (when nobody was living there), discrepancy. Vacant properties are at higher risk for owner-caused damage (arson) – an occupancy of "Vacant" at time of loss is a known red flag.
- **home_security_system**: If present, it's suspicious if a burglary claim shows no alarm trigger. If false burglary is claimed and they have an alarm that never went off or wasn't enabled, investigators will ask why.
- **appraised_value** vs claimed losses: If someone claims \$500k in contents lost but the entire home is worth \$300k, that's unlikely – either they overinsured contents or are exaggerating. It gives a scale.
- This dimension also allows grouping claims by property: if a single property has multiple fire claims over years, that pattern emerges (possible serial arson or someone milking insurance repeatedly).
- **construction_year/type** can indicate if damage should have certain characteristics (old wiring might cause accidental fires – but if a brand new house "just happened" to have an electrical fire, one might scrutinize more for intentional causes).

Property Policy Dimension

Column Name	Data Type	Description
policy_id	String (UUID)	Policy identifier for property insurance (primary key).
policyholder_id	String (UUID)	Foreign key to Insured (who holds this policy).
property_id	String (UUID)	Foreign key to Property (the property insured under this policy).
policy_start_date	Date	Policy start date.
policy_end_date	Date	Policy end/renewal date.
coverage_types	String	Coverage included (Dwelling, Personal Property/Contents, Liability, Loss of Use, etc.).
coverage_limits	String	Key limits (e.g., dwelling coverage amount, contents limit, deductible).
premium_annual	Float	Annual premium for the policy.
last_claim_date	Date	Date of last claim on this policy (if any).
flags	String	*Optional:* Notes such as "Increased Coverage Recently", "Previous Cancellation for Non-pay", etc.

Rationale: The Property Policy dimension links the insured, the property, and coverage details. It helps catch timing and coverage-related fraud indicators:
- The link between **property_id** and policy means we know exactly what property is covered. If a claim is filed for a property not actually insured or after a policy lapsed, that's immediately flagged.
- **coverage_limits** on dwelling and contents are critical. Fraudulent claims often aim for near these limits [30]. If someone schedules (specifically insures) very high-value items and then a theft occurs, that's suspicious timing. If the **flags** show "Increased Coverage Recently", and then a huge loss happens, that aligns with known patterns (e.g., insuring jewelry then claiming theft soon after).
- **last_claim_date** can show if claims are frequent. If last_claim_date is very recent and another claim happens, it could be either bad luck or fraud (some might claim multiple small incidents or one big one in parts).
- Also, policy attributes combined with property: if **policyholder_id** has multiple property policies with claims (say, multiple properties burning over years under the same owner), that cross-property pattern is visible.
- Premium and coverage might indirectly highlight something: very low premium (meaning lower coverage) but extremely high claimed loss could indicate exaggeration beyond what anyone would reasonably insure vs. risk.

Overall, the Policy dimension ensures any analysis of claims can factor in whether the policy was new, had changes, or the coverage context, which all feed into fraud rules.

Claim Fact (Property)

Column Name	Data Type	Description
claim_id	String (UUID)	Unique property claim ID (primary key).
policy_id	String (UUID)	Foreign key to Property Policy (which policy is claiming).
insured_id	String (UUID)	Foreign key to Insured (who filed the claim).
property_id	String (UUID)	Foreign key to Property (the property involved in claim).
incident_date	Date	Date the loss incident occurred (or was noticed).
claim_report_date	Date	Date the claim was reported to insurer.
loss_type	String	Type of loss event (Fire, Theft, Water Damage, Wind/Hail, Liability, etc.).
loss_cause	String	Cause of loss as reported (e.g., “kitchen grease fire”, “forced entry burglary”, “pipe burst”).
claimed_amount_dwelling	Float	Amount claimed for building/structural damage (if applicable).
claimed_amount_contents	Float	Amount claimed for contents (personal property loss).
claimed_amount_other	Float	Other claimed costs (ALE – Additional Living Expenses, etc.).
claim_status	String	Status of claim (Open, Closed, Denied, Under Investigation).
suspicious_flags	String	*Optional:* Flags (e.g., “Large Loss after Recent Policy Change”, “No Police Report for Theft”).

Rationale: The property Claim fact provides a summary of the claim at the policy level. Important fields and their fraud implications:
- **incident_date vs claim_report_date**: A long delay in reporting (especially for theft or damage) is unusual – honest insureds report promptly. A delay might indicate fabricated or inflated claims (perhaps waiting to gather fake receipts, etc.).
- **loss_type and loss_cause**: Key for consistency checks. If loss_type is Fire, but loss_cause is something odd or not matching evidence (like claiming an “electrical fire” but investigators find signs of accelerant), that discrepancy would emerge in investigation, though not purely from data – however capturing cause allows comparing with known red flag causes. Certain causes are inherently more suspicious (fire, theft) and might trigger rule-based flags automatically because they are common fraud avenues [23] .
- **claimed_amount_dwelling vs coverage**: If someone claims the full rebuild of house (e.g., a fire total loss), we check if it equals policy limit. Repeated cases of total losses could indicate fraud. If the claimed amounts

are just under limits across dwelling/contents, again suggests careful padding [30] .
- **claimed_amount_contents**: This is where exaggeration often happens (inflating value of items). By comparing with an inventory (below) and coverage, we can spot if it's maxed out or unrealistic relative to the household.
- Breaking down claim into dwelling vs contents vs ALE is useful because fraudsters might overload one category. For example, claiming maximum Additional Living Expenses without proper receipts might be a trick – the data model can show if ALE is claimed at all, and a rule could flag if ALE claimed but damage was minor (why would they need living expenses?).
- **loss_cause** and **loss_type** also allow linking to external data: e.g., if cause is "hail storm on X date", the insurer can verify if a storm occurred at that location/date. If not, that's a blatant red flag. Similarly, if theft is claimed, one expects a police report; a flag in suspicious_flags might note "No Police Report" if data from the claimant indicates they didn't file one.
- The **suspicious_flags** can catch things like if an adjuster notes inconsistencies or system auto-flags like prior losses, recent insurance changes, etc.

This claim fact ties to detailed sub-tables for deeper investigation (repair estimates, contents inventory, etc.), enabling both high-level analytics (e.g., total claimed vs paid, frequency of claims by cause) and fine-grained checks.

Contractor Repair Estimate Fact (Property)

Column Name	Data Type	Description
estimate_id	String (UUID)	Unique ID for a repair/rebuild estimate line (primary key).
claim_id	String (UUID)	Foreign key to Claim Fact (claim for which repair is estimated).
component	String	Repair component or damaged area (e.g., "Roof replacement", "Kitchen cabinets", "Plumbing repair").
damage_description	String	Description of damage and needed work (e.g., "roof shingles missing, need replacement on 50% area").
contractor_id	String (UUID)	Identifier for contractor/firm giving the estimate (could link to a Contractor Dimension).
estimated_cost	Float	Estimated cost for this component repair.
materials_cost	Float	Sub-breakdown: estimated materials cost (if provided).
labor_cost	Float	Sub-breakdown: estimated labor cost.
final_approved_cost	Float	Final approved cost for this component after insurance review.

Rationale: The Contractor Estimate table itemizes building repair costs, similar in spirit to the auto repair table but for property. This is vital to detect **inflated repair costs or contractor fraud**:
- **component** and **damage_description** allow cross-checking the scope of repairs with the cause of loss. If a claim is for a small kitchen fire but the components include unrelated areas (like "repaint entire house"), that indicates over-claiming.

- Having the **contractor_id** lets the insurer see if the same contractor is repeatedly involved in large claims, possibly coaching insureds or part of a fraud ring. NICB has noted patterns of certain contractors or public adjusters in many questionable claims [31]. If one contractor's name appears in an abnormally high number of fire claims in a region, big warning sign.
- **estimated_cost vs final_approved_cost**: Large discrepancies mean the adjuster reduced a likely inflated quote. Tracking this over time by contractor can identify serial over-estimators.
- **materials_cost vs labor_cost**: If provided, these can show if a contractor is claiming very high labor hours or expensive materials. E.g., using top-tier materials unnecessarily (marble instead of standard counters in repairs) to inflate cost could be flagged. Or claiming labor that seems excessive (100 hours to paint one room).
- Also, if **estimated_cost** for certain components far exceed typical benchmarks (insurers have data for average cost to, say, replace a roof of X size), the system can flag it.
- If an insured declines to provide multiple contractor bids or only uses their "preferred" contractor who gave the highest bid, that context might come from adjuster notes but the presence of only one estimate can be noted. In a more complex model we might track multiple estimates, but here perhaps one per claim for simplicity, or if multiple contractors, multiple entries with same claim_id. That too can be telling if one is very high and one much lower.

In short, detailed repair lines prevent a lump-sum claim from hiding exaggeration. Each line can be scrutinized for necessity and cost reasonableness, which is at the core of combating padded property claims.

Contents Inventory Fact (Property)

Column Name	Data Type	Description
item_id	String (UUID)	Unique ID for a claimed item (primary key).
claim_id	String (UUID)	Foreign key to Claim Fact (claim for which item loss is claimed).
item_name	String	Name/description of item (e.g., "65-inch Samsung TV", "Diamond Ring").
category	String	Category of item (Electronics, Jewelry, Furniture, etc.).
claimed_value	Float	Value claimed for this item (what insured says it was worth).
quantity	Integer	Quantity of this item lost (e.g., 3 for three similar chairs).
ownership_proof	String	Proof provided (Receipt, Photo, None).
item_age_years	Integer	Age of the item in years (if known or estimated).
covered_flag	Boolean	Whether item was specifically covered/scheduled (true if listed on policy separately, otherwise standard coverage).

Rationale: In theft or content-damage claims, this table lists each item the insured says was lost or destroyed. It's essential for uncovering **inflated or fake contents claims**:
- **item_name, category, claimed_value:** This allows the insurer to judge if values are inflated. If someone

lists a “Generic Laptop” at \$5000, that’s likely inflated. The adjuster or system can cross-reference typical prices. A rule might flag any single item above a certain value without specific proof.
- **quantity:** Watch for improbably high counts (e.g., “10 leather jackets” for one person, could be fake).
- **ownership_proof:** If None provided for high-value items, that’s a red flag. Legit expensive items usually have receipts or appraisals. A rule can flag high-value jewelry with no proof as suspect. Conversely, if proof is provided, system could later verify it (fake receipts are possible, but lack of any proof is more straightforwardly suspicious).
- **item_age_years:** If an item’s age is high but claimed as if new, there might be exaggeration of value (since actual cash value would depreciate). Some policies cover replacement cost, but still, claiming brand-new for an antique item is a tactic. This field helps adjusters apply proper depreciation or spot if someone claimed an old item was new to maximize payout.
- **covered_flag:** If an item is scheduled (specifically insured with appraisal, common for jewelry or art), then claiming it is straightforward. If not, and it’s very high value, might exceed normal sub-limits (e.g., jewelry typically has a limit if not scheduled). A fraudster might lie about multiple expensive pieces hoping insurer doesn’t notice sub-limit. The system can catch that – e.g., total jewelry claimed vs policy jewelry limit. If far above and not scheduled, either they won’t get paid fully (which they might not realize) or it’s a tactic to pressure insurer. In any case, unusual unscheduled high-value items are suspicious.

By analyzing the entire inventory list: are the items consistent with the insured’s profile and income? (Not explicitly stored here, but e.g., someone with a modest home claiming a grand piano, several luxury watches, etc.). Patterns like claiming all items were top-of-line brand new is a red flag (the **category** field could be used to see if they disproportionately claim high-end electronics, etc.).

The inventory data also helps in aggregate: if one insured claimed two 65-inch TVs in one theft, and two years later another theft claim with two 65-inch TVs again, that pattern emerges (via prior_claim_count or just linking claims by insured and comparing items).

Photo Evidence Fact (Property)

Column Name	Data Type	Description
photo_id	String (UUID)	Unique ID for a photo document (primary key).
claim_id	String (UUID)	Foreign key to Claim Fact (claim this evidence relates to).
photo_type	String	Type of photo (Damage Scene, Item Proof, Receipts, etc.).
description	String	Description of content (e.g., “Living room fire damage”, “Remains of item”).
file_path	String	Storage path/URL of the image file.
taken_date	DateTime	When the photo was taken (if available).
verified_flag	Boolean	Flag if adjuster verified the photo’s authenticity/ relevance.

Rationale: Similar to auto, photo evidence for property claims is stored. Types might include pictures of the damage (burned interior, broken door for break-in, etc.) or even pre-loss photos of items (if insured supplies any), receipts images, etc. Fraud implications:
- If a burglary claim has **no photos of forced entry** (and none provided), that's suspicious. Conversely, if they provide a photo of a broken door, adjusters will check if it looks fresh or staged. Having the photo linked means an investigator can quickly retrieve and examine it.
- For fire claims, photos of the aftermath can sometimes reveal irregular patterns (like suspicious burn patterns indicating accelerant). While that analysis is manual or done by experts, the system can note if those experts flagged something (reflected perhaps in verified_flag or an investigator's notes).
- **taken_date** for receipts or item photos: if all receipts photos are from after the loss, they might have quickly fabricated them. Or if item photos are stock images from internet (adjuster might mark not verified).
- Also, ensuring that photos correspond to the correct property (e.g., correct address visible or matching the property's features) is something a human verifies, but by cataloging them, we ensure no evidence is just kept offline.
- If the **same photo of damage** were ever reused in another claim (like two unrelated insureds showing the same kitchen fire pic, which has happened in fraud rings), storing the images with metadata allows for forensic analysis or hash matching to catch that.

Essentially, the photo evidence table ensures a **complete dataset** for the claim is available for analysis. Even if automated flags on photos are limited, including them supports the investigative workflow and any potential AI image analysis for fraud in the future.

Payment Fact (Property)

Column Name	Data Type	Description
payment_id	String (UUID)	Unique payment ID (primary key).
claim_id	String (UUID)	Foreign key to Claim Fact (claim this payment is for).
payee	String	Who was paid (e.g., insured, mortgage lender, contractor).
payee_id	String	Identifier of payee if applicable (could be insured_id, contractor_id, etc.).
payment_date	Date	Date of disbursement.
amount_paid	Float	Amount paid in this transaction.
payment_purpose	String	Purpose (Dwelling repair payout, Contents settlement, ALE reimbursement, etc.).
method	String	Payment method (Check, Electronic, etc.).
final_settlement_flag	Boolean	True if this payment completes the claim settlement (full and final).

Rationale: Payment records for property claims are similar to auto. They help in detecting anomalies like:
- **payee**: If large payments go to a contractor directly, was that authorized? If a contractor is paid directly

many times (especially one that also gave high estimates), that's notable. Or if the insured insists payments be made out to a strange third party, big red flag.
- If the insured has a mortgage, often large dwelling repair checks include the bank as a payee. If our data shows payment went solely to insured when normally a mortgagee is involved, could indicate something amiss (maybe they lied about having no mortgage to get money directly, which is more a compliance issue but could be tied to fraud intent).
- **amount_paid vs claimed**: if a claim was inflated, likely amount_paid will be significantly less (after adjuster cuts) or claim could be denied partially. Patterns where claimed_amount is always much higher than amount_paid for an insured might indicate they consistently exaggerate (insurer always negotiates them down).
- **final_settlement_flag**: This can be used in situations where multiple payments happen (advance, then final, etc.). If multiple partial payments occur over time unusually (like incremental payouts), it might be because the insured kept finding "new" damages – a tactic to extend a claim artificially. Tracking if a claim had many disbursements might highlight those that start small but balloon (suspicious possibly).
- As always, the timeline matters: quick payouts on large claims, or any irregular payment patterns (like paying out even before an adjuster report is in) could indicate internal fraud or procedural lapses that fraudsters exploit.

By capturing payments, we also enable linking to external recovery efforts (like subrogation or fraud recoveries not directly asked here, but the data is ready). Importantly, if a fraud scheme involves funneling money to certain accounts, having payee and payee_id data makes financial analysis possible (for example, a corrupt insured and contractor could actually be the same person via different entities – comparing addresses or IDs might reveal it).

Alignment with Rule-Based and ML/DL Fraud Detection

The above data model is designed to align with real-world best practices for fraud detection, supporting both **rule-based systems** and **Machine Learning/Deep Learning (ML/DL)** approaches:

- **Rule-Based Detection:** Industry organizations (like the NICB) publish fraud indicators and red flags [32], and insurers implement business rules to flag claims with those characteristics. Our model explicitly captures the data needed for these rules. For example, we record timing details (policy start vs claim, incident vs report delay), relationships (links between claims, people, vehicles, providers), and granular cost breakdowns – all of which feed rules such as "multiple claims from same insured in short time" or "claim amount just below coverage limit". Each potential red flag identified in the **Fraud Factors** section can be operationalized: e.g., a rule can query the **Treatment Fact** for high-frequency upcoded procedures, or the **Incident/Claim** tables for mismatches and patterns (like repeated accidents at the same location). By having separate tables for each aspect (visits, labs, repair lines, inventory items, etc.), the system can isolate anomalies at a fine-grained level (like an item with no proof, or a procedure with no corresponding diagnosis). These tables act as a **single source of truth** where rules engines can easily pull required fields without missing data. The inclusion of fields such as `suspicious_flags` in claim tables also allows rule engines to record triggers for audit tracking.

- **Machine Learning & Deep Learning:** Modern fraud detection increasingly relies on ML/DL models to identify subtle patterns across large datasets. Our model stores data at a very **granular level (lowest possible level of granularity) following dimensional standards** [28], which is ideal for

feature engineering. Every dimension and fact can provide features: demographics of claimants, historical behavior (e.g., prior_claim_count), text descriptions of incidents (which NLP models could analyze for inconsistent narratives), network linkages (e.g., common addresses or providers indicating collusion), and so on. The **rich data model** ensures that if a pattern exists (like providers frequently co-occurring with certain claimants, or an insured always using the same contractor), it's recorded in a structured way that ML models can exploit. For instance, a graph-based ML model could use the relationships between Insured, Vehicles, and Accident participants to spot rings. A deep learning model might look at sequences of claims or the combination of multiple features (amounts, times, codes) to predict fraud – all these inputs are available in our schema.

- **Real-World Standards and Completeness:** The model incorporates tables for all relevant entities (payments, evidence, etc.), providing a **complete audit trail for each claim**. This completeness is not only good for compliance and investigation, but also for machine learning – models can be trained on full history including outcomes (fraud found or not) with all supporting evidence. In practice, such comprehensive data capture is needed; as noted in an industry context, storing detailed claims data including **individual payments and changes** aligns with best practices for an insurance data warehouse [29]. Our model mirrors this, capturing each payment and detail. This level of detail allows advanced analytics and AI to thrive. For example, a supervised ML model can learn from historical fraud cases by looking at dozens of features (e.g., number of lab tests in claim, provider's average cost, insured's claim frequency, policy age, etc.) – our design provides those features readily. Moreover, unsupervised techniques (anomaly detection) can be applied on numeric fields like costs, counts of services, time intervals, etc., thanks to the structured format.

- **Flexibility for Analytical Techniques:** Because the data is organized dimensionally, it's optimized for querying and aggregating – beneficial for both rule engines (which might query counts and sums) and ML (which might require aggregating data by group for feature creation). The model supports generating features like "average cost per claim for this provider" (by aggregating Treatment costs by provider), or "number of claims this insured had in past 5 years" (by querying Claim facts by insured_id). These can be done on the fly or materialized as needed. It's designed to be **fast and flexible for insurance-specific queries** [33], which is crucial when running fraud analytics across millions of records.

- **Network and Graph Analysis:** Real-world fraud detection often leverages network analysis (graph-based features) to uncover rings and collusion. Our model inherently supports that by separating entities and linking them via keys. For example, a graph of relationships can be built from the data: patients to providers (via claims), drivers to vehicles to accidents, contractors to homeowners, etc. Entities like phone numbers or addresses (present in Person/Provider dimensions) further connect the graph. This means our data can be fed to graph algorithms to detect communities of frequently connected nodes (potential fraud rings). It's essentially **future-proofed** for more complex analytics beyond traditional rules.

- **Transparency and Explainability:** A critical aspect of fraud detection, especially with ML, is explaining why a claim was flagged. Since our model keeps the raw factors (rather than, say, burying them in one big free-form record), an explainable AI or a human investigator can pinpoint the reason: e.g., "Claim was flagged because patient had 5 expensive lab tests (Lab Fact) that don't match the diagnosis (Claim/Diagnosis) and provider has history of upcoding (Provider stats from Treatment Fact)". Each piece of that explanation is a direct data point in our model, traceable back to

source tables. This traceability was a goal, as indicated by maintaining a **complete audit trail for machine learning, rules, and investigations** [34] – every alert can be backed by data from one of the tables (like an entry in suspicious_flags or a count from a fact table).

- **Compliance and Integration:** The schema aligns with what insurers use in practice (fact/dimension models in data warehouses), making it easier to integrate real data and third-party sources. For example, integration with external fraud databases (like NICB's Questionable Claims database) is facilitated by having clearly defined entities (we could easily tag a provider or insured if they appear in an external blacklist). Rule-based systems used in insurance (like business rule engines) typically require structured input as we've prepared. Meanwhile, ML pipelines can directly pull from these tables or create feature views. The model's design acknowledges both approaches: rules may catch known patterns first (fast and precise for known fraud typologies), while ML can sift through the remainder for unknown patterns – together improving fraud detection coverage [35] [36].

In conclusion, this data model provides a robust foundation for fraud detection. It is **comprehensive**, capturing all relevant aspects of claims (from medical procedures and lab results to car repair details and property inventories), and **aligned with real-world standards** (dimension modeling, granular data, complete records). It enables a spectrum of detection methods – from simple flag rules (e.g., "flag if >3 claims in year" using prior_claim_count) to complex AI algorithms – ensuring the insurer can effectively identify and investigate fraudulent claims while maintaining evidence and context for each case.

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